

Knowledge Distillation for Efficient On-Device Image Classification

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Problem Statement

The deployment gap

- Large CNNs (ResNet-110: **1.7M params**) achieve high accuracy.
- But edge devices have hard limits on memory, latency, and power
- Model Compression for NN introduced in 2006
- Train a compact *student* to mimic a large pretrained *teacher*

Student trained on Hard Labels

- CCE gradient: $\partial L / \partial z_k^S = \hat{y}_k^S - y_k$
- Non-target classes always have $y_k=0$, pushing all wrong logits to $-\infty$ regardless of class similarity

Teacher's soft output encodes *dark knowledge*

Class	Hard label	Soft ($\tau=4$)
dog	1.00	0.71
wolf	0.00	0.14
fox	0.00	0.09
cat	0.00	0.04
other 96	0.00	0.02

Dark Knowledge [Hinton 2015]

Nonzero probabilities on wrong classes encode inter-class similarity that hard labels discard.

Methodology: Response-Based Knowledge Distillation

Temperature-scaled softmax

$$\hat{y}_k^{T,\tau} = \frac{\exp(z_k^T / \tau)}{\sum_j \exp(z_j^T / \tau)}$$

$\tau > 1$ flattens distribution $\Rightarrow$ amplifies non-target gradients

Distillation objective

$$L_{\text{KD}} = \alpha L_{\text{hard}} + (1 - \alpha) \tau^2 L_{\text{soft}}$$

τ^2 compensates gradient shrinkage:

$$\partial L_{\text{soft}} / \partial z_k^S \propto 1 / \tau^2$$

Hyperparameter sweeps

- Temperature: $\tau \in \{2, 4, 8, 16\}$
- Loss weight: $\alpha \in \{0.1, 0.5, 0.9\}$

Capacity study

- *Depth*: ResNet- $\{20, 32, 44, 56\}$, width 16
- *Width*: ResNet-20, multiplier $w \in \{0.5, 1, 2\}$
- Shared pivot: ResNet-20 ($w=1$)

Training config

Setting	Value
Optimiser	SGD + Momentum (0.9)
LR / decay	0.1 / $\times 0.1$ at ep. 60, 80
Weight decay	5×10^{-4}
Batch / Epochs	128 / 100 (student)

Dataset & Experimental Setup

CIFAR-100

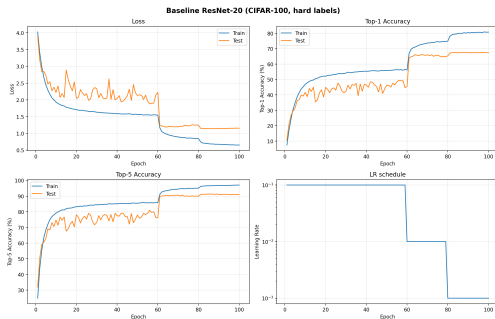
- 60,000 images, 32×32 , 100 classes
- 50k train / 10k test (500 per class)

Models (best-checkpoint)

	Params	Top-1
Teacher (RN-110)	1.74M	73.48%
Baseline (RN-20)	0.28M	67.63%

Gap = 5.85 pp = distillation headroom

Baseline ResNet-20 training on CIFAR-100 (100 epochs).



LR decays at epochs 60 & 80 produce the visible accuracy jumps. Train–test gap ≈ 12.5 pp indicates mild overfitting.

Results: Hyperparameter Sweeps & Capacity Scaling

Temperature ($\alpha=0.5$)

τ	Top-1	Δ
2	69.49%	+1.86
4	69.64%	+2.01
8	69.98%	+2.35
16	69.73%	+2.10

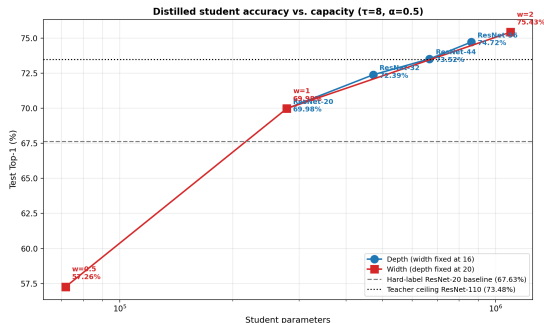
Inverted-U; $\tau^*=8$ (Hinton: $\tau \approx 4$)

Loss weight ($\tau=8$)

α	Top-1	Δ
0.1	69.74%	+2.11
0.5	69.98%	+2.35
0.9	69.07%	+1.44

$\alpha=0.9$ recovers only 62% of peak gain; soft labels provide genuinely new information

Capacity scaling ($\tau=8, \alpha=0.5$)

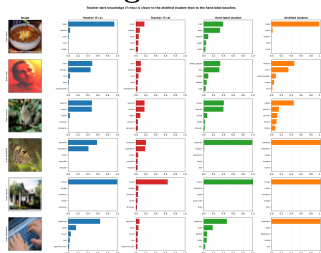


3 of 5 students exceed teacher (73.48%).

Best: ResNet-20 $w=2$ reaches **75.43%** (+7.80 pp), still 1.6 $\times$ smaller. Low-capacity failure: $w=0.5$ falls **10.37 pp** below baseline.

Dark Knowledge & Calibration Analysis

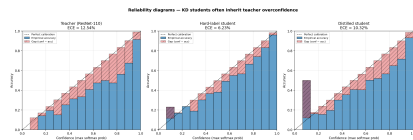
Why soft labels help: dark knowledge



Each row: image | teacher $\tau=1$ | teacher $\tau=4$ | hard-label student | distilled student.
At $\tau=4$, probability spreads to similar classes; distilled student mimics this structure while hard-label student does not.

Side-effect: calibration degrades

Model	Top-1	ECE↓
Teacher	73.48%	12.54%
Hard-label (base)	67.63%	6.23%
Distilled ($\tau=8$)	69.98%	10.32%
Distilled ($w=2$)	75.43%	9.97%



KL term transfers teacher's **overconfidence profile**, not just the argmax. Hard-label student is best calibrated despite lowest Top-1.
Post-hoc calibration required for probability-sensitive applications.